

Hugging Face Hub

Model, Datasets, Spaces, and Much More



Hugging Face

Sergio Paniego ([@sergiopaniego](https://twitter.com/sergiopaniego))

Machine Learning Engineers @ Hugging Face

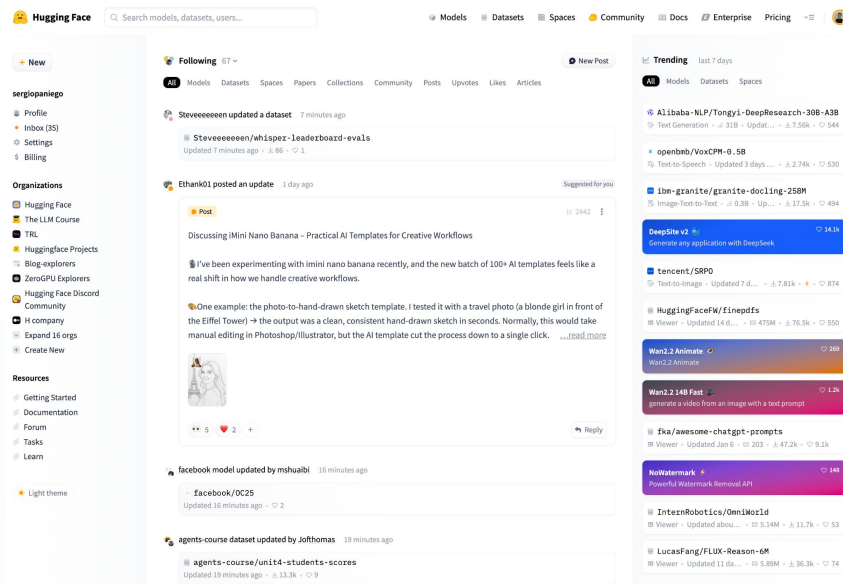
What is Hugging Face 🤗

AI open platform for building, sharing and deploying AI (HF Hub):

- Models, datasets, apps...
- Posts, blogs, courses, papers, leaderboards...

2.8M
public models

900K+
public datasets



1M+
daily downloads

1M+
daily visitors

What is Hugging Face 🤗

Set of open source projects to empower AI development and advancement



Transformers



Datasets



Trackio



Hub



30+

Libraries



Transformers.js



Diffusers

timm



Accelerate



PEFT



smolagents

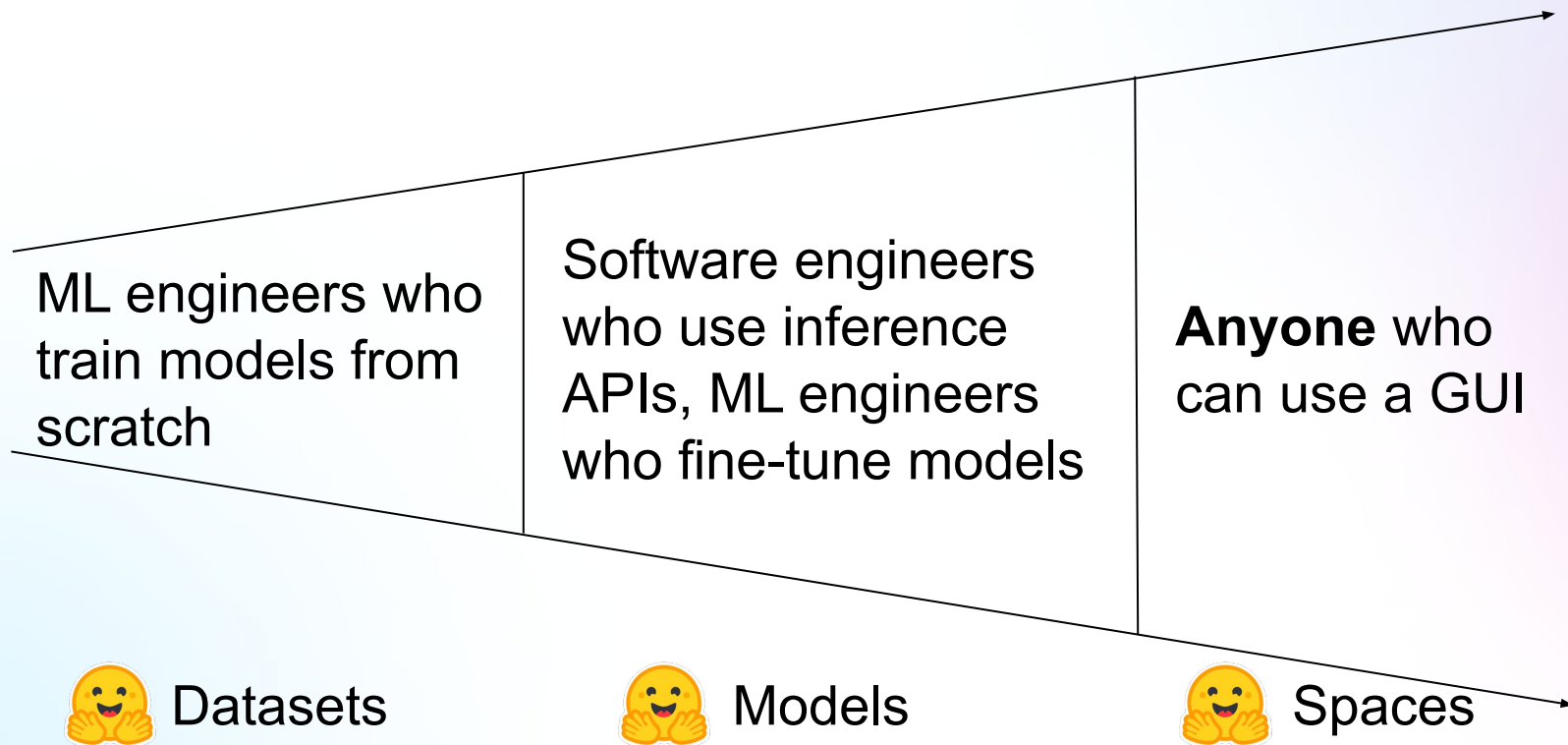


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Open vs Closed

	Open	Closed / Proprietary
Security	Models can be self-hosted , data stays in your environment	Models cannot be self-hosted. Data is sent outside your environment to vendor
Control	The timing and nature of updates are controlled by you	Updates and changes to performance can happen without notice
Customization	Full source code access to customize the model for your needs	Limited ability to customize for your needs
Transparency	Inspect code and data provides better auditability and understandability	No ability to audit or understand performance
Cost	Typical lower long term cost due to smaller model size	larger model size and proprietary premium often balanced by decreased cost from server-side optimization
Latency	Lower latency due to on premise and smaller model sizes	Often greater latency due to larger model sizes + API latency
Quality	No single approach is best. Each use case will vary. Proprietary is typically closer to the frontier of performance .	
Examples	  OpenAI  Meta  deepseek  MISTRAL AI  Microsoft	 OpenAI  ANTHROPIC

How HF democratizes ML/AI 🤝



Used everywhere in the AI world

15,000+ startups and enterprises



Open source contributors



Hardware partners



Cloud partners



On-prem partners



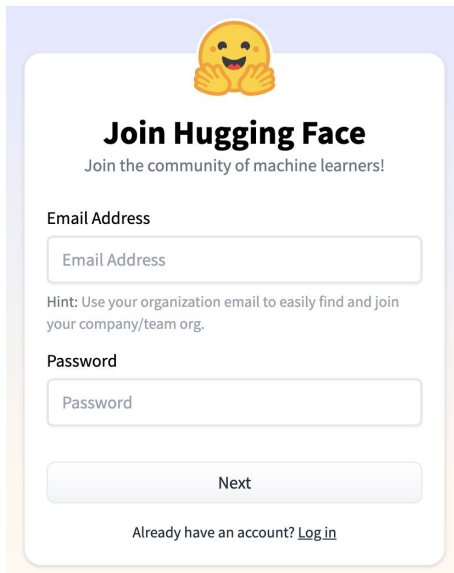
Hugging Face



Creating a HF Hub account

Let's create an account and explore the possibilities:

- Go to hf.co/join

A screenshot of the Hugging Face registration page. At the top is a yellow emoji with its hands clasped. Below it, the text "Join Hugging Face" is in bold, followed by "Join the community of machine learners!". There are two input fields: "Email Address" and "Password", each with a placeholder of the same name. A hint below the email field says "Hint: Use your organization email to easily find and join your company/team org.". Below the password field is a "Next" button. At the bottom, it says "Already have an account? [Log in](#)".



Join Hugging Face

Join the community of machine learners!

Email Address

Hint: Use your organization email to easily find and join your company/team org.

Password

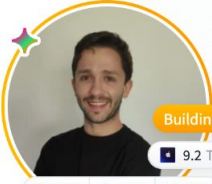
Next

Already have an account? [Log in](#)

HF Hub profile

 **Hugging Face**

[Models](#) [Datasets](#) [Spaces](#) [Buckets](#) [NEW](#) [Docs](#) [Enterprise](#) [Pricing](#) [⌵](#) 



Building on HF
9.2 TFLOPS

222 172 108

Sergio Paniego PRO
sergiopaniego 

🌐 1,610 followers · 107 following

[+ New](#) [Edit profile](#) [Settings](#)

<https://sergiopaniego.github.io/>
[X sergiopaniego](#) [sergiopaniego](#)
[in sergio-paniego-blanco](#)

 **AI & ML interests**
None yet

 **Recent Activity**
Updated dataset ...

Posts 84

Post

Great experience yesterday at PyTorch Conf Europe in Paris 🇫🇷

196 ...read more

[View all Posts](#)

Articles 16

Article

Welcome Gemma 4: Frontier multimodal intelligence on device

826 ...

[View all Articles](#)

Collections 9

Bringing Autonomous Driving RL to OpenEnv and TRL resources

Blog: <https://huggingface.co/blog/sergiopaniego/bringing-carla-to-openenv-trl/>

CARLA Environment Server
Control a Carla driving simulator by sending actions

CARLA Environment Server
Control a CARLA driving simulator with custom actions

Carla Grpo Trolley
Visualize your program's I/O activity in real time

Research & Long-Form Blog Posts

In-depth technical articles and research pieces published by Hugging Face

The Ultra-Scale Playbook
The ultimate guide to training LLM on large GPU Clusters
3.77k

The Smol Training Playbook
The secrets to building world-class LLMs
3.1k

Evaluation Guidebook
Explore LLM benchmark trends over time
300

First Steps: Open Data in NLP with ...
221

 **Hugging Face**

Exploring models

Going to Models and exploring trending/tasks/libraries...

The screenshot displays the Hugging Face website's 'Models' section. The top navigation bar includes links for Models, Datasets, Spaces, Buckets, Docs, Enterprise, and Pricing. A search bar is located at the top left. The left sidebar contains filters for Tasks, Libraries, Parameters, Apps, and Inference Providers. The main content area shows a list of models, including 'zai-org/GLM-5.1', 'google/gemma-4-31B-it', 'openbmb/VoxCPM2', 'dealignai/Gemma-4-31B-JANG_4M-CRACK', 'MiniMaxAI/MiniMax-M2.7', 'netflix/void-model', 'k2-fsa/OmniVoice', 'google/gemma-4-E4B-it', 'Jackrong/Qwen3.5-27B-Claude-4.6-Opus-Reasoning-Dist..', 'google/gemma-4-26B-A4B-it', 'unsloth/gemma-4-26B-A4B-it-GGUF', 'LilaRest/gemma-4-31B-it-NVFP4-turbo', 'HauhausCS/Gemma-4-E4B-Uncensored-HauhausCS-Aggressive', 'nvidia/Gemma-4-31B-IT-NVFP4', 'google/gemma-4-E2B-it', 'tencent/HY-Embodied-0.5', 'prism-ml/Bonsai-8B-gguf', and 'baidu/Qianfan-OCR'. Each model card displays its name, task, parameters, and other relevant information.

Hugging Face Search models, datasets, users...

Models 2,782,428 Filter by name

Full-text search Inference Available 11 Sort: Trending

Tasks

- Text Generation
- Any-to-Any
- Image-Text-to-Text
- Image-to-Text
- Image-to-Image
- Text-to-Image
- Text-to-Video
- Text-to-Speech
- +44

Parameters

1B 6B 12B 32B 128B >500B

Libraries

- PyTorch
- TensorFlow
- JAX
- Transformers
- Diffusers
- sentence-transformers
- Safetensors
- ONNX
- GGUF
- Transformers.js
- MLX
- +42

Apps

- vLLM
- llama.cpp
- MLX LLM
- LM Studio
- Ollama
- Jan
- Draw Things
- +7

Inference Providers

- Groq
- Novita
- Cerebras
- SambaNova
- Nscale
- fal
- Hyperbolic
- Together AI
- +16

Models

- zai-org/GLM-5.1**
Text Generation · 754B · Updated 1 day ago · 28.8k · 1.09k
- google/gemma-4-31B-it**
Image-Text-to-Text · 33B · Updated 3 days ago · 2.24M · 1.79k
- openbmb/VoxCPM2**
Text-to-Speech · Updated 5 days ago · 7.45k · 763
- dealignai/Gemma-4-31B-JANG_4M-CRACK**
Image-Text-to-Text · 6B · Updated 3 days ago · 99.1k · 964
- MiniMaxAI/MiniMax-M2.7**
Text Generation · 229B · Updated 1 day ago · 873 · 549
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- unsloth/gemma-4-26B-A4B-it-GGUF**
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Image-Text-to-Text · 4B · Updated 3 days ago · 582 · 136
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Text Generation · 8B · Updated 6 days ago · 74.4k · 568
- baidu/Qianfan-OCR**
Image-Text-to-Text · 5B · Updated 18 days ago · 44.8k · 1.14k

Hugging Face

Exploring models

Examples of models 🖥️:

- [Gemma 4 \(Google\)](#)
- [Qwen 3.5 \(Qwen\)](#)
- [Llama 4 \(Meta\)](#)
- [GLM-5.1 \(zai\)](#)
- [MiniMax M2.7 \(Minimax\)](#)
- [gpt-oss \(OpenAI\)](#)
- [Flux.2 \(Black Forest\)](#)
- [Gemma 4 GGUF \(Unsloth\)](#)
- [DeepSeek R1 \(DeepSeek\)](#)
- [LFM2.5 \(LiquidAI\)](#)
- [Nemotron 3 \(nvidia\)](#)



Gemma 4



deepseek



cohere



Meta



Exploring datasets

Datasets

The screenshot displays the Hugging Face Datasets interface. The top navigation bar includes links for Models, Datasets, Spaces, Posts, Docs, Enterprise, and Pricing. A search bar is located on the left side of the top bar. The left sidebar contains filters for Modalities (3D, Audio, Geospatial, Image, Tabular, Text, Time-series, Video), Size (rows) (a slider from <1K to >1T), and Format (json, csv, parquet, imagefolder, soundfolder, webdataset, text, arrow). The main content area shows a grid of dataset cards, each with a title, a brief description, and statistics such as the number of rows, size, and likes. The cards are sorted by trending.

Hugging Face Search models, datasets, users...

Models Datasets Spaces Posts Docs Enterprise Pricing

Datasets 366,444 Filter by name Full-text search Sort: Trending

Modalities

3D Audio Geospatial Image Tabular Text Time-series Video

Size (rows)

<1K >1T

Format

json csv parquet imagefolder soundfolder webdataset text arrow

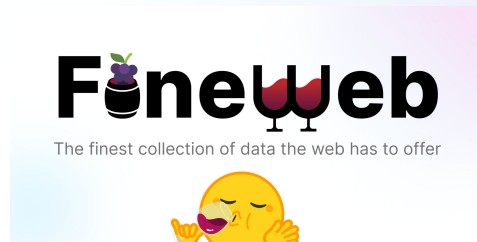
Dataset Cards:

- zwhe99/DeepMath-103K**
Viewer · Updated 5 days ago · 103k · 9.8k · 137
- Anthropic/values-in-the-wild**
Preview · Updated 3 days ago · 185 · 85
- nvidia/OpenCodeReasoning**
Viewer · Updated 8 days ago · 753k · 11.9k · 283
- OpenGVLab/InternVL-Data**
Preview · Updated about 11 hours ago · 634 · 42
- openai/mcrr**
Viewer · Updated 9 days ago · 2.4k · 2.73k · 125
- fka/awesome-chatgpt-prompts**
Viewer · Updated Jan 6 · 203 · 11.5k · 7.72k
- future-technologies/Universal-Transformers-Dataset**
Viewer · Updated 8 days ago · 70.1M · 4.57k · 72
- nvidia/ClimbLab**
Viewer · Updated 2 days ago · 1.1B · 9.83k · 28
- nvidia/Llama-Nemotron-Post-Training-Dataset**
Viewer · Updated 7 days ago · 3.91M · 7.19k · 423
- bh2821/LightNovel5000**
Viewer · Updated 7 days ago · 6.12M · 646 · 24

Exploring datasets

Example of datasets 📁:

- [FineWeb \(Hugging Face\)](#)
- [HH-RLHF \(Anthropic\)](#)
- [Wikipedia \(Wikimedia\)](#)
- [The Cauldron \(Hugging Face\)](#)
- [GSM8K \(openAI\)](#)



ANTHROPIC



Exploring Spaces

AI apps! 📱

Spaces · The AI App Directory

+ New Space What is Spaces?

Ask anything you want to do with AI

Image Generation Video Generation Text Generation Language Translation Speech Synthesis 3D Modeling Object Detection Text Analysis Image Editing Code Generation Question Answering Data Visualization Voice Cloning

Spaces of the week 21 Apr 2025 Filter by name Sort: Relevance

Dia 1.6B 🗣️ Generate audio from text using Nari TTS nari-labs 1 day ago 334	Chat with Bitnet-b1.58-2B-4T 🗣️ Chat with Microsoft's 1.58bit Bitnet model! suayptalha 3 days ago 27	Vevo for Zero-shot VC, TTS, and More 🗣️ Controllable Zero-Shot Voice Imitation amphion about 6 hours ago 27	MESA 🗣️ Text-to-terrain model (reflectance and elevation) mikonvergence 7 days ago 16
MotionShop2 🎬 Replace characters in a video with characters in photos 3DAIGC 8 days ago 27	DetailGen3D 🎮 Generate detailed 3D models from reference images and coarse models VAST-AI 5 days ago 29	FLUX.1 Dev ControlNet Union Pro 2.0 🔥 Create images using prompts and control images Shakker-Labs 6 days ago 42	AudioX 🗣️ Generate audio and video from text prompts Zeyue7 7 days ago 30

Exploring Spaces

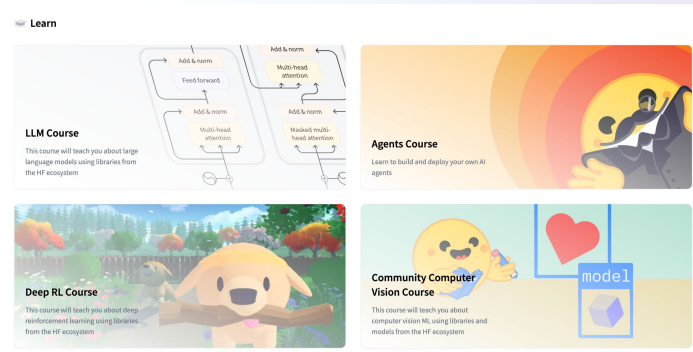
HF Spaces

Some examples 📱 :

- [DeepSite](#): generates websites using DeepSeek
- [LFM2-MoE](#): talk with LFM2-MoE
- [InstantCoder](#): generated apps using Gemini
- [OminiControl Art](#): transform images into artistic styles
- [Voxtral TTS](#): Voxtral model example
- [Kolors-Virtual-Try-On](#): try clothes on people
- [Kimodo](#): generate high-quality motions from text prompts

Education-Community

- [Learn](#)
- [Blog](#)
 - [Research & Long-Form Posts](#)
- [Community articles](#)
- [Papers](#)
- Benchmarks
 - [SWE-bench](#)
 - [MMLU](#)
- [Discord](#)



How do we build with it?

Find model → Customize → Deploy



Find the model

 **Hugging Face**

Models | Datasets | Spaces | Buckets NEW | Docs | Enterprise | Pricing | 

Main | Tasks | Libraries | Languages | Licenses | Other

Tasks

Text Generation | Any-to-Any | Image-Text-to-Text | Image-to-Text | Image-to-Image | Text-to-Image | Text-to-Video | Text-to-Speech | +44

Parameters

<1B | 6B | 12B | 32B | 128B | >500B

Libraries

PyTorch | TensorFlow | JAX | Transformers | Diffusers | sentence-transformers | Safetensors | ONNX | GGUF | Transformers.js | MLX | +42

Apps

vLLM | llama.cpp | MLX LM | LM Studio | Ollama | Jan | Draw Things | +7

Inference Providers

Groq | Novita | Cerebras | SambaNova | Nscale | fal | Hyperbolic | Together AI | +16

Models 2,782,428  Filter by name

Full-text search  Inference Available 11 Sort: Trending

 **zai-org/GLM-5.1**
 Text Generation · ∙ 754B · Updated 1 day ago · ∙ 28.8k · ∙ 1.09k

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 Image-Text-to-Text · ∙ 5B · Updated 18 days ago · ∙ 44.8k · ∙ 1.14k

Use it with 🥳 Transformers

The Transformers Library: standardizing model definitions

Published May 15, 2025

[Update on GitHub](#)



Lysandre
lysandre

Follow



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pcuenca

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julien-c

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TLDR: Going forward, we're aiming for Transformers to be the pivot across frameworks: if a model architecture is supported by transformers, you can expect it to be supported in the rest of the ecosystem.

Transformers was created in 2019, shortly following the release of the BERT Transformer model. Since then, we've continuously aimed to add state-of-the-art architectures, initially focused on NLP, then growing to Audio and computer vision. Today, transformers is the default library for LLMs and VLMs in the Python ecosystem.





Use it with 🤗 Transformers



```
from transformers import pipeline

# Load the text generation pipeline with Qwen3
pipe = pipeline(
    "text-generation",
    model="Qwen/Qwen3-0.6B",
    dtype="auto",
    device_map="auto"
)

# Define the conversation (messages)
messages = [
    {
        "role": "user",
        "content": "Give me a short introduction to large language model."
    }
]

# Generate the response
outputs = pipe(
    messages,
    max_new_tokens=512,
)

# The pipeline returns structured output
print(outputs[0]["generated_text"][-1]["content"])
...
<think>
Okay, the user wants a short introduction to a large language model. Let me start by ...
</think>
```

A large language model (LLM) is a type of artificial intelligence that can understand, generate, and respond to human language. These models are trained on massive datasets to understand complex topics and produce creative, context-aware content. They are used in various applications, from writing essays to providing translations and answering questions.



Customize with

TRL allows you to fine-tune VLMs using different methods: SFT, DPO, MPO, GRPO, GSPO, RLOO, and Online DPO.
PEFT and quantization support

```
from trl import SFTTrainer
from datasets import load_dataset

trainer = SFTTrainer(
    model="Qwen/Qwen3-0.6B",
    train_dataset=load_dataset("trl-lib/Capybara", split="train"),
)
trainer.train()
```



Deploy the model

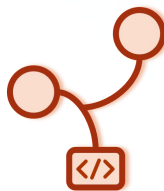
Inference and serving frameworks

Inference Providers (TGI)

Hugging Face Endpoints

 vLLM

LLaMA 



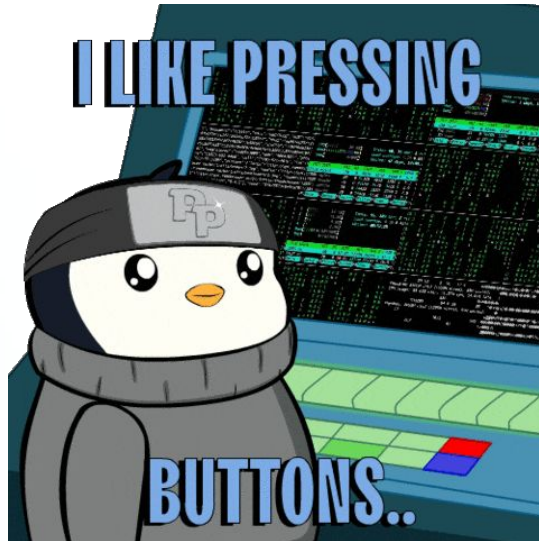
 SGL



Let's get to code! 🧑💻

Let's use HF in code!

<https://colab.research.google.com/drive/1d5Xopgr6Vgg9nJqcRP2TrSGpxwYWA-Cu?usp=sharing>



Let's get to code! 🧑💻

We'll review some key functionalities of HF

- Pipeline
- Transformers
 - Download and use model
- Datasets
 - Download and access dataset
- Train model combining both
- Deploy trained model to Gradio app in HF (ZeroGPU)
- See where to find more examples!



Thank you

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Machine Learning Engineers @ Hugging Face